Module 3: Two-Dimensional Defect Diffusion-Reaction Evolution and Annealing

Derivation-Focused Physics Note for `RadiationEffects_proj2`

Time evolution of radiation-induced defects in silicon after the initial damage event

Purpose of this note. This note develops the defect-evolution module that turns an initial radiation damage map into a time-dependent defect landscape inside silicon. The goal is to derive, from conservation of defect number and constitutive transport ideas, the two-dimensional diffusion-reaction equations used to model defect migration, recombination, clustering, dissociation, and annealing. This module provides the evolving damage state that later feeds the electrostatic, thermal, and carrier-transport parts of the project.

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Symbols and acronyms used in this document

Symbol	Meaning	Typical units
C	Defect concentration field	m^{-3}
C_i	Concentration of defect species i	m^{-3}
$\mathbf{J}_C$	Defect flux	$\mathrm{m}^{-2}\ \mathrm{s}^{-1}$
D	Defect diffusivity	m^2/s
D_0	Diffusion prefactor, typically $a^2\nu_D/(2d)$ in a jump-diffusion picture	m^2/s
R	Net local reaction term	$\mathrm{m}^{-3}\ \mathrm{s}^{-1}$
k_{ann}	First-order annealing rate constant	s^{-1}
k_0	Annealing prefactor or attempt rate for the modeled annealing channel	s^{-1}
T	Temperature field	K
E_D	Effective migration activation energy for defect motion between metastable lattice configurations	J or eV
E_{ann}	Effective activation energy for defect removal or transformation during annealing	J or eV
k_B	Boltzmann constant	J/K
x, y	Cartesian spatial coordinates	m
t	Time	s
L	Characteristic length scale	m
τ_D	Diffusion timescale	s
τ_{ann}	Annealing timescale	s
$M(t)$	Total defect inventory	dimension depends on dimensional reduction

Symbol	Meaning	Typical units
$\hat{\mathbf{n}}$	Outward unit normal vector	dimensionless

1 Purpose of Module 3 and its role in the project

Module 1 determines where radiation deposits energy. That deposited energy is the initial physical insult, but it is not yet the same thing as long-term damage. The material consequences of a radiation event evolve afterward. Defects can diffuse, recombine, cluster, dissociate, or anneal. Module 3 is the part of the project that turns the initial damage map into an evolving defect landscape.

This module is central because many downstream device consequences depend not only on how much damage was created, but also on where that damage migrates, how quickly it disappears, and whether it becomes more complex. Module 2 uses the defect population to modify space charge. Module 4 uses temperature to change mobility and reaction rates. Module 5 uses the evolving defect landscape to alter recombination and trapping. Thus Module 3 is the kinetic heart of the entire framework.

2 Conservation-law starting point

2.1 Defect concentration as a continuum field

Let $C(x, y, t)$ denote the concentration of a defect species, in units of number per volume. Even though defects are discrete lattice-scale objects, at the device scale it is useful to represent them by a coarse-grained concentration field. This is the same continuum logic used throughout transport theory: a sufficiently large but locally small averaging volume contains many defects, so a smooth field description becomes meaningful.

2.2 Balance law for a defect species

Consider a small material element. The concentration inside that element can change for three reasons:

1. defects can flow into or out of the element,
2. defects can be created inside the element,
3. defects can be destroyed or transformed inside the element.

The local conservation or balance law is therefore

$$\frac{\partial C}{\partial t} + \nabla \cdot \mathbf{J}_C = R(C, T, \ldots), \tag{2.1}$$

where $\mathbf{J}_C$ is the defect flux and R is the local net reaction term. The right-hand side is positive when reactions create the species and negative when reactions consume it.

This equation is the natural defect analogue of the continuity equation. The key modeling task is therefore to specify the flux law and the reaction law.

3 Diffusive transport from Fick's law

3.1 Why diffusion appears

If the defect concentration is spatially nonuniform, there is a thermodynamic tendency for the distribution to smooth out. Regions of high concentration tend to feed regions of low concentration through random migration. At the continuum level, the simplest constitutive relation for that process is Fick's law:

$$\mathbf{J}_C = -D\nabla C. \tag{3.1}$$

The minus sign is crucial. It says the flux points down the concentration gradient. Defects move from more concentrated regions toward less concentrated regions.

3.2 From Fick's law to the diffusion equation

Substituting Fick's law into the balance law gives

$$\frac{\partial C}{\partial t} + \nabla \cdot (-D\nabla C) = R, \tag{3.2}$$

so

$$\frac{\partial C}{\partial t} = \nabla \cdot (D\nabla C) + R. \tag{3.3}$$

If D is spatially uniform, this becomes

$$\frac{\partial C}{\partial t} = D\nabla^2 C + R. \tag{3.4}$$

This is the diffusion-reaction equation. It is the core governing equation for Module 3.

3.3 Physical meaning of the divergence form

The divergence form,

$$\frac{\partial C}{\partial t} = \nabla \cdot (D \nabla C) + R, \tag{3.5}$$

is more fundamental than the constant-D Laplacian form because it remains correct even when D varies in space or depends on temperature. The diffusion operator is not merely "curvature." It is the divergence of a flux. That distinction becomes essential once Module 4 temperature fields are coupled in and the diffusivity varies from place to place.

4 Reaction and annealing terms

4.1 Why a reaction term is needed

Radiation damage is not frozen in place. A vacancy may meet an interstitial and recombine. A mobile point defect may be absorbed at a sink. A defect complex may form from simpler constituents. Some defects may dissociate. These processes are not represented by diffusion alone. They require a local source-sink term.

Therefore the reaction term R is not optional. It is the place where defect chemistry and defect kinetics enter the model.

4.2 First-order annealing as the simplest reduced model

The simplest physically meaningful reaction model is first-order annealing,

$$R = -k_{\text{ann}} C, \tag{4.1}$$

where k_{ann} is the annealing rate constant. The governing equation then becomes

$$\frac{\partial C}{\partial t} = \nabla \cdot (D \nabla C) - k_{\text{ann}} C. \tag{4.2}$$

This says that defects spread by diffusion while being removed locally at a rate proportional to the current concentration.

4.3 Why first-order annealing is useful

First-order annealing is not the most general reaction law, but it is a highly useful reduced model because:

1. it gives a clean analytic limit when diffusion is absent,

2. it captures the idea of an intrinsic decay timescale,
3. it is easy to verify numerically,
4. it provides a baseline before introducing more complicated coupled kinetics.

4.4 General multi-species reaction structure

For multiple defect species C_i, the balance law becomes

$$\frac{\partial C_i}{\partial t} = \nabla \cdot (D_i \nabla C_i) + R_i(C_1, C_2, \ldots, T). \tag{4.3}$$

This is the more general architecture the project can grow into. In the earliest implementation, however, a single effective defect species is sufficient to establish the transport framework.

5 Reduction to the two-dimensional project form

The project has adopted a 2D architecture for Modules 2–6. Thus the defect concentration becomes

$$C = C(x, y, t). \tag{5.1}$$

For Cartesian coordinates,

$$\nabla \cdot (D \nabla C) = \frac{\partial}{\partial x}\left(D \frac{\partial C}{\partial x}\right) + \frac{\partial}{\partial y}\left(D \frac{\partial C}{\partial y}\right). \tag{5.2}$$

Therefore the 2D governing equation is

$$\frac{\partial C}{\partial t} = \frac{\partial}{\partial x}\left(D \frac{\partial C}{\partial x}\right) + \frac{\partial}{\partial y}\left(D \frac{\partial C}{\partial y}\right) + R(C, T, \ldots). \tag{5.3}$$

If the diffusivity is uniform and the reaction is first-order annealing,

$$\frac{\partial C}{\partial t} = D\left(\frac{\partial^2 C}{\partial x^2} + \frac{\partial^2 C}{\partial y^2}\right) - k_{\mathrm{ann}} C. \tag{5.4}$$

This is the baseline project equation for Module 3.

6 Initial conditions from radiation damage maps

6.1 How Module 1 feeds Module 3

Geant4 does not directly output an already-evolved defect concentration. It outputs deposited energy, event locations, and possibly a binned damage map. Module 3 must translate that radiation information into an initial defect condition or source term.

The simplest reduced mapping is

$$C(x, y, 0) = C_0(x, y), \tag{6.1}$$

where C_0 is obtained from a damage-generation rule based on the Geant4 deposition map. In a time-resolved variant one could instead use a source term $S(x, y, t)$, but for a first post-irradiation evolution model it is enough to treat radiation as an initial condition.

6.2 Why this distinction matters

This point is conceptually important: Geant4 determines where the initial insult occurs, while Module 3 determines what that insult becomes afterward. Conflating these two would erase the very defect-kinetics physics the project is trying to capture.

7 Boundary conditions and their physical meaning

7.1 Zero-flux boundary condition

The most natural default condition for a closed region with no defect exchange across the boundary is

$$\hat{\mathbf{n}} \cdot \mathbf{J}_C = 0. \tag{7.1}$$

Using Fick's law, this becomes

$$\hat{\mathbf{n}} \cdot \nabla C = 0. \tag{7.2}$$

This means there is no net defect flow across the boundary. Physically, the boundary acts as a reflecting boundary or symmetry plane.

7.2 Dirichlet boundary condition

Another possibility is to prescribe the concentration directly:

$$C = C_b \quad \text{on a boundary.} \tag{7.3}$$

This is less common for a closed silicon device region but can represent contact with a reservoir or an imposed sink environment.

7.3 Reactive or sink boundary condition

A more physical future refinement is a Robin-type condition that represents surface absorption or sink behavior,

$$-\hat{\mathbf{n}} \cdot D\nabla C = h_s(C - C_{\mathrm{eq}}). \tag{7.4}$$

This says the boundary can absorb or emit defects depending on deviation from an equilibrium value. Such a condition is useful when surfaces or interfaces act as strong sinks for mobile defects.

8 Analytic limiting cases used for verification

8.1 Pure annealing with no diffusion

If diffusion is switched off, $D = 0$, and the reaction is first-order,

$$\frac{\partial C}{\partial t} = -k_{\mathrm{ann}} C. \tag{8.1}$$

The solution is

$$C(t) = C(0)e^{-k_{\mathrm{ann}} t}. \tag{8.2}$$

If the initial state is spatially uniform, the entire field should remain spatially uniform while decaying exponentially. This is one of the cleanest verification tests because the exact solution is known.

8.2 Pure diffusion with no reaction

If $R = 0$, the equation becomes

$$\frac{\partial C}{\partial t} = D\nabla^2 C. \tag{8.3}$$

Under zero-flux boundaries, the total amount of defect,

$$M(t) = \int C \, \mathrm{d}V, \tag{8.4}$$

should be conserved. The concentration profile should smooth and broaden, but the integral should remain constant.

8.3 Uniform-state preservation

If the initial condition is exactly uniform and the boundaries are consistent with that uniformity, then diffusion should generate no structure. A correct numerical scheme should preserve the uniform state up to roundoff.

9 Characteristic timescales and regime interpretation

9.1 Diffusion timescale

Let L be a characteristic length scale. The diffusive relaxation time is approximately

$$\tau_D \sim \frac{L^2}{D}. \tag{9.1}$$

This says diffusion becomes slower for larger structures and faster for larger diffusivity.

9.2 Annealing timescale

For first-order annealing, the intrinsic decay time is

$$\tau_{\text{ann}} \sim \frac{1}{k_{\text{ann}}}. \tag{9.2}$$

9.3 Competition between diffusion and annealing

The ratio

$$\frac{\tau_D}{\tau_{\text{ann}}} \sim \frac{L^2 k_{\text{ann}}}{D} \tag{9.3}$$

indicates which process dominates.

- If $\tau_D \ll \tau_{\text{ann}}$, diffusion redistributes defects before much annealing occurs.
- If $\tau_D \gg \tau_{\text{ann}}$, defects disappear locally before they can migrate far.

This competition is central to the interpretation of results. It determines whether damage remains localized, spreads out, or fades quickly.

10 Temperature coupling and Arrhenius physics

A major reason Module 4 exists is that both diffusion and annealing are temperature sensitive. The compact Arrhenius formulas used in Module 3 are not arbitrary fitting expressions. They come from the idea that a thermally activated process must overcome an energy barrier before it can proceed.

10.1 Why thermal activation produces an exponential

Suppose a defect process requires access to a higher-energy configuration separated from the initial state by an energy barrier E_a. In thermal equilibrium, the probability of finding a system with

energy E is weighted by the Boltzmann factor

$$P(E) \propto \exp\left(-\frac{E}{k_B T}\right). \tag{10.1}$$

Therefore, the fraction of defects with enough thermal energy to reach a transition state at energy barrier E_a scales like

$$P(\text{barrier crossed}) \propto \exp\left(-\frac{E_a}{k_B T}\right). \tag{10.2}$$

If the process also has an attempt frequency or attempt rate ν_0, then the resulting rate takes the generic activated form

$$\Gamma(T) = \nu_0 \exp\left(-\frac{E_a}{k_B T}\right). \tag{10.3}$$

This is the physical origin of Arrhenius behavior. The exponential is the thermal penalty for climbing an energy barrier, while the prefactor measures how often the system attempts to cross that barrier.

10.2 Derivation of the diffusion Arrhenius law

For point-defect migration in a crystal, diffusion can be viewed as a sequence of thermally activated jumps between neighboring lattice sites. Let a be a characteristic jump distance and let Γ_D be the jump frequency. For an unbiased random walk, the diffusivity scales as jump length squared times jump frequency. Up to an order-unity geometric factor,

$$D \sim a^2 \Gamma_D. \tag{10.4}$$

More explicitly, for a simple random walk in d spatial dimensions,

$$D = \frac{a^2 \Gamma_D}{2d}. \tag{10.5}$$

If each jump requires the defect to overcome a migration barrier E_D, then the jump frequency is thermally activated:

$$\Gamma_D(T) = \nu_D \exp\left(-\frac{E_D}{k_B T}\right), \tag{10.6}$$

where ν_D is an attempt frequency associated with lattice vibrations or local configurational attempts. Substituting this into the random-walk expression gives

$$D(T) = \frac{a^2 \nu_D}{2d} \exp\left(-\frac{E_D}{k_B T}\right). \tag{10.7}$$

The prefactor is commonly grouped into a single constant,

$$D_0 \equiv \frac{a^2 \nu_D}{2d}, \tag{10.8}$$

so that

$$D(T) = D_0 \exp\left(-\frac{E_D}{k_B T}\right). \tag{10.9}$$

This is the diffusion Arrhenius law used in Module 3.

The quantity E_D is therefore not just a symbol inserted by hand. It is the effective *migration activation energy*: the energy barrier that a defect must overcome to move from one metastable lattice configuration to another. In a more microscopic picture, E_D may contain contributions from the local crystal environment, strain, charge state, and nearby impurities or defects. In the reduced continuum model, all of that microscopic complexity is compressed into the single effective barrier E_D and prefactor D_0.

10.3 Derivation of the annealing Arrhenius law

Annealing is also a thermally activated process, but its physical meaning differs from diffusion. Annealing represents the disappearance or transformation of a defect population through processes such as recombination, structural rearrangement, dissociation of an unstable complex, or conversion into a lower-energy configuration.

Suppose a defect-removal channel requires crossing an activation barrier E_{ann}. Then the microscopic rate per defect takes the same activated form,

$$\Gamma_{\text{ann}}(T) = \nu_{\text{ann}} \exp\left(-\frac{E_{\text{ann}}}{k_B T}\right). \tag{10.10}$$

If the annealing channel is first order, the number of defects removed per unit volume per unit time is proportional to the local concentration:

$$R_{\text{ann}} = -\Gamma_{\text{ann}}(T)\, C. \tag{10.11}$$

We then identify the first-order annealing constant as

$$k_{\text{ann}}(T) \equiv \Gamma_{\text{ann}}(T) = k_0 \exp\left(-\frac{E_{\text{ann}}}{k_B T}\right), \tag{10.12}$$

with

$$k_0 \equiv \nu_{\text{ann}}. \tag{10.13}$$

Thus the first-order decay law

$$\frac{\partial C}{\partial t} = -k_{\text{ann}}(T) C \tag{10.14}$$

comes directly from the assumption that each defect has an independent thermally activated probability per unit time of being removed.

The quantity E_{ann} is the effective *annealing activation energy*. Physically, it is the barrier associated

with the dominant defect-removal or defect-transformation pathway being represented in the reduced model. It need not be identical to E_D, because migration and annealing are not the same microscopic event. A defect may move relatively easily yet require a larger barrier to recombine, or vice versa.

10.4 Why the prefactors and activation energies are called effective

In real irradiated silicon, there is usually not just one defect species and not just one microscopic pathway. Different vacancies, interstitials, complexes, and charge states can each have different migration barriers and reaction barriers. If the continuum model uses a single species concentration C and a single pair of Arrhenius parameters, then

$$D_0,\ E_D,\ k_0,\ E_{\text{ann}} \tag{10.15}$$

should be interpreted as *effective parameters* summarizing the dominant kinetics over the temperature and timescale range of interest. They may come from atomistic calculations, experiments, or calibration to measured annealing curves.

10.5 Module 3 equation with thermal coupling

Once Module 4 supplies $T(x, y, t)$, the Module 3 equation becomes

$$\frac{\partial C}{\partial t} = \nabla \cdot (D(T)\nabla C) - k_{\text{ann}}(T)C. \tag{10.16}$$

If D varies spatially through its temperature dependence, then the diffusion term can be expanded as

$$\nabla \cdot (D(T)\nabla C) = D(T)\nabla^2 C + \nabla D(T) \cdot \nabla C. \tag{10.17}$$

The first term is the familiar Laplacian smoothing term. The second term appears because temperature gradients create spatial gradients in diffusivity, which in turn bias how quickly different regions relax.

In 2D Cartesian form, the coupled equation is

$$\frac{\partial C}{\partial t} = \frac{\partial}{\partial x}\left(D(T)\frac{\partial C}{\partial x}\right) + \frac{\partial}{\partial y}\left(D(T)\frac{\partial C}{\partial y}\right) - k_{\text{ann}}(T)C. \tag{10.18}$$

Now the thermal field changes not only how quickly defects move, but also how quickly they disappear. Higher temperature generally increases both diffusivity and annealing rate because the Boltzmann penalty for barrier crossing becomes less severe.

This coupling is one of the strongest reasons the project must be multiphysics. Defect kinetics and temperature cannot be treated as unrelated after the architecture has matured.

11 How Module 3 couples to other modules

11.1 Input from Module 1

Module 1 supplies the initial damage map or source distribution.

11.2 Output to Module 2

Module 3 provides the evolving concentrations $C_i(x, y, t)$ that are converted into defect charge density in Module 2.

11.3 Input from Module 4

Module 4 provides the temperature field that changes diffusivities and reaction rates.

11.4 Output to Module 5

Module 5 uses the defect distribution to model trap-assisted recombination, trapping, and lifetime degradation.

Thus Module 3 is not isolated. It is the evolving material-state layer that communicates with all of the main downstream device-physics modules.

12 What this module captures and what it does not

Module 3 captures continuum-scale migration and reaction of defects. It does not directly resolve atomistic trajectories or explicitly compute formation energies from first principles. Those deeper physics can be folded into the continuum parameters D, R, and their temperature dependence, but the module itself is a device-scale defect-kinetics model rather than a molecular-dynamics or density-functional-theory solver.

This distinction is useful because it keeps the project scope realistic while still preserving physically meaningful links to microscopic input data.

13 Summary of the final project equation

The baseline 2D Module 3 equation is

$$\frac{\partial C}{\partial t} = \frac{\partial}{\partial x}\left(D\frac{\partial C}{\partial x}\right) + \frac{\partial}{\partial y}\left(D\frac{\partial C}{\partial y}\right) + R(C, T, \ldots). \tag{13.1}$$

For the first reduced implementation,

$$\frac{\partial C}{\partial t} = D\left(\frac{\partial^2 C}{\partial x^2} + \frac{\partial^2 C}{\partial y^2}\right) - k_{\mathrm{ann}} C. \tag{13.2}$$

With temperature coupling,

$$\frac{\partial C}{\partial t} = \nabla \cdot (D(T)\nabla C) - k_{\mathrm{ann}}(T) C. \tag{13.3}$$

This is the equation set that turns a radiation-created damage map into an evolving defect landscape suitable for coupling to electrostatics, thermal physics, and carrier transport.